

TCX2101: Calculus and Linear Algebra for Computing

Chapter 6: Matrix Algebra

Introduction to Matrices

Definition

Definition

A (real-valued) matrix is a rectangular array of (real) numbers

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} = (a_{ij})_{m \times n} = (a_{ij})_{i=1}^m {}_{j=1}^n,$$

where $a_{ij} \in \mathbb{R}$ are real numbers. The size of the matrix is said to be $m \times n$ (read as m by n), where m is the number of rows and n is the number of columns.

The numbers in the array are called entries. The (i, j) -entry, a_{ij} , $i = 1, \dots, m$, $j = 1, \dots, n$, is the number in the i -th row j -th column.

Question

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & -1 & 5 & 10 \\ 3 & 5 & 13 & 23 & 41 \\ -7 & 2 & 0 & 0 & 11 \end{pmatrix}$$

1. What is the **size** of the matrix?

3×5

2. What is the **(2,3)-entry**?

13

Remarks

- ▶ The size of a matrix is read as m by n . One should not multiply the numbers. For example,

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & -1 & 5 & 10 \\ 3 & 5 & 13 & 23 & 41 \\ -7 & 2 & 0 & 0 & 11 \end{pmatrix} \text{ is a 3 by 5 matrix, not a size 15 matrix.}$$

- ▶ There is a need to add a comma when labelling the (i,j) -entry, $a_{i,j}$, if there is ambiguity. For example, a_{123} may mean the $(1,23)$ -entry, in which case we label it as $a_{1,23}$ instead, or the $(12,3)$ -entry, in which case we label it as $a_{12,3}$ instead.

Example

We may define a matrix by some formula on its entry. For example,

$$\mathbf{A} = (a_{ij})_{2 \times 3}, \quad a_{ij} = i + j$$

$$\mathbf{A} = \begin{pmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{pmatrix}$$

Example

$$\mathbf{B} = (b_{ij})_{3 \times 2}, b_{ij} = (-1)^{i+j}$$

$$\mathbf{B} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \\ 1 & -1 \end{pmatrix}$$

Vectors

A $n \times 1$ matrix is called a (column) vector, and a $1 \times n$ matrix is called a (row) vector.

Remark

If it is not specified whether the vector is a column or a row vector, by default we will assume it is a column vector.

Zero Matrices

All entries equal 0, denoted as $\mathbf{0}_{m \times n}$. Not necessarily a square matrix.

Example

$$\mathbf{0}_{2 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{0}_{3 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{0}_{4 \times 1} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{0}_{1 \times 1} = (0).$$

Square Matrices

Number of rows = number of columns

$$\mathbf{A} = (a_{ij})_n = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix},$$

Example

$$\begin{pmatrix} 2 & -3 \\ 7 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 1 & 0 & 0 \\ -1 & -1 & 0 & 1 \end{pmatrix}.$$

Square Matrices

Definition

- i. A size $n \times n$ matrix is a square matrix of order n .
- ii. The entries a_{ii} , $i = 1, 2, \dots, n$, (explicitly, $a_{11}, a_{22}, \dots, a_{nn}$) are called the diagonal entries of the (square) matrix.

Diagonal, Scalar, Identity matrices

1. Diagonal matrix $\mathbf{D} = (a_{ij})_n$, $a_{ij} = 0$ for $i \neq j$. Denote as

$$\mathbf{D} = \text{diag}(d_1, d_2, \dots, d_n) = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}.$$

2. Scalar matrix $\mathbf{C} = (a_{ij})$, $a_{ij} = \begin{cases} c & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$, $\mathbf{C} = \text{diag}(c, c, \dots, c) = \begin{pmatrix} c & 0 & \cdots & 0 \\ 0 & c & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & c \end{pmatrix}.$

3. Identity matrix $\mathbf{I} = (a_{ij})$, $a_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$, $\mathbf{I}_n = \text{diag}(1, 1, \dots, 1) = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$

A scalar matrix can also be denoted as $\mathbf{C} = c\mathbf{I}$, where $\mathbf{I}$ is the identity matrix. See later for definition of scalar multiplication.

Question

Which of the following options are true? The 3 by 3 zero matrix $\mathbf{0}_{3 \times 3}$ is a

- (a) diagonal matrix.
- (b) scalar matrix.
- (c) identity matrix.

Triangular Matrices

Upper triangular $\mathbf{A} = (a_{ij})$, $a_{ij} = 0$ for all $i > j$:

$$\begin{pmatrix} * & * & \cdots & * \\ 0 & * & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & * \end{pmatrix}$$

Strictly upper triangular $\mathbf{A} = (a_{ij})$, $a_{ij} = 0$ for all $i \geq j$:

$$\begin{pmatrix} 0 & * & \cdots & * \\ 0 & 0 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}$$

Lower triangular $\mathbf{A} = (a_{ij})$, $a_{ij} = 0$ for all $i < j$:

$$\begin{pmatrix} * & 0 & \cdots & 0 \\ * & * & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \cdots & * \end{pmatrix}$$

Strictly lower triangular $\mathbf{A} = (a_{ij})$, $a_{ij} = 0$ for all $i \leq j$:

$$\begin{pmatrix} 0 & 0 & \cdots & 0 \\ * & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \cdots & 0 \end{pmatrix}$$

Question

Which of the following statements are true?

- (i) Every square matrix is row equivalent to an upper triangular matrix.

- (ii) Every square matrix is row equivalent to a strictly upper triangular matrix.

Symmetric Matrices

$$\mathbf{A} = (a_{ij})_n, \quad a_{ij} = a_{ji}.$$

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{pmatrix}$$

Example

$$\begin{pmatrix} 1 & 2 & b \\ a & 0 & 4 \\ 3 & c & d \end{pmatrix} \text{ is symmetric } \Leftrightarrow a = 2, b = 3, c = 4, d \in \mathbb{R}.$$

Question

A symmetric upper triangular matrix is a zero matrix. True or false?

Matrix Algebra

Equality

Two matrices are equal if they have the same size and their corresponding entries are equal;

$\mathbf{A} = (a_{ij})_{n \times m}$ and $\mathbf{B} = (b_{ij})_{k \times l}$ are equal if and only if $n = k$, $m = l$, and $a_{ij} = b_{ij}$ for all $i = 1, \dots, n$, $j = 1, \dots, m$.

Example

1. $\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \neq \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$ for any $a, b, c, d, e, f \in \mathbb{R}$ since the matrices do not have the same sizes.
2. $\begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Leftrightarrow a = b = 1, c = 3, d = 2.$

Matrix Addition and Scalar Multiplication

Definition

1. Scalar multiplication: $c \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} = \begin{pmatrix} ca_{11} & ca_{12} & \cdots & ca_{1n} \\ ca_{21} & ca_{22} & \cdots & ca_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{m1} & ca_{m2} & \cdots & ca_{mn} \end{pmatrix}.$

2. Matrix addition:

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{pmatrix} = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{pmatrix}$$

Examples

$$1. \ 5 \begin{pmatrix} 6 & 1 & -1 \\ 2 & -4 & 3 \\ 4 & 9 & -11 \end{pmatrix} = \begin{pmatrix} 30 & 5 & -5 \\ 10 & -20 & 15 \\ 20 & 45 & -55 \end{pmatrix}$$

$$2. \ \begin{pmatrix} 1 & 2 & -3 \\ 4 & 5 & 6 \end{pmatrix} + \begin{pmatrix} 2 & 1 & 5 \\ 1 & -3 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 3 & 2 \\ 5 & 2 & 8 \end{pmatrix}$$

Remark

1. Matrix addition is only defined between matrices of the same size.
2. $-\mathbf{A} = (-1)\mathbf{A}$.
3. Matrix subtraction is defined to be the addition of a negative multiple of another matrix,

$$\mathbf{A} - \mathbf{B} = \mathbf{A} + (-1)\mathbf{B}.$$

Properties of Matrix Addition and Scalar Multiplication

Theorem

For matrices $\mathbf{A} = (a_{ij})_{m \times n}$, $\mathbf{B} = (b_{ij})_{m \times n}$, $\mathbf{C} = (c_{ij})_{m \times n}$, and real numbers $a, b \in \mathbb{R}$,

- (i) (Commutative) $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$,
- (ii) (Associative) $\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}$,
- (iii) (Additive identity) $\mathbf{0}_{m \times n} + \mathbf{A} = \mathbf{A}$,
- (iv) (Additive inverse) $\mathbf{A} + (-\mathbf{A}) = \mathbf{0}_{m \times n}$,
- (v) (Distributive law) $a(\mathbf{A} + \mathbf{B}) = a\mathbf{A} + a\mathbf{B}$,
- (vi) (Scalar addition) $(a + b)\mathbf{A} = a\mathbf{A} + b\mathbf{A}$,
- (vii) (Associative) $(ab)\mathbf{A} = a(b\mathbf{A})$,
- (viii) If $a\mathbf{A} = \mathbf{0}_{m \times n}$, then either $a = 0$ or $\mathbf{A} = \mathbf{0}$.

Remark

By the associativity of matrix addition, if $\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_k$ are k matrices of the same size, we simply write the sum as

$$\mathbf{A}_1 + \mathbf{A}_2 + \cdots + \mathbf{A}_k,$$

without the parenthesis (brackets).

Matrix Multiplication

$$\mathbf{AB} = (a_{ij})_{m \times p} (b_{ij})_{p \times n} = (\sum_{k=1}^p a_{ik} b_{kj})_{m \times n}$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{pmatrix}$$
$$= \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} & a_{11}b_{13} + a_{12}b_{23} + a_{13}b_{33} & a_{11}b_{14} + a_{12}b_{24} + a_{13}b_{34} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} & a_{21}b_{13} + a_{22}b_{23} + a_{23}b_{33} & a_{21}b_{14} + a_{22}b_{24} + a_{23}b_{34} \end{pmatrix}$$

That is, the (i, j) -entry of the product $\mathbf{AB}$ is the sum of the product of the entries in the i -th row of $\mathbf{A}$ with the j -th column of $\mathbf{B}$.

Example

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 3 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} (1)(1) + (2)(2) + (3)(-1) & (1)(1) + (2)(3) + (3)(-2) \\ (4)(1) + (5)(2) + (6)(-1) & (4)(1) + (5)(3) + (6)(-2) \end{pmatrix}$$
$$= \begin{pmatrix} 2 & 1 \\ 8 & 7 \end{pmatrix}$$

Question

- What is the $(2,3)$ -entry of the product

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ -1 & 2 & 1 & 1 \\ 0 & 1 & 2 & -1 \\ 3 & 1 & 0 & 1 \\ 2 & -5 & 9 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & -1 & 0 & 0 \\ 3 & 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}?$$

- What is the $(2,3)$ -entry of the product

$$\begin{pmatrix} 1 & 2 & 3 \\ -1 & 0 & 9 \end{pmatrix} \begin{pmatrix} 1 & 0 & 5 \\ 1 & 3 & -2 \end{pmatrix}?$$

Remark

1. For $\mathbf{AB}$ to be defined, the number of columns of $\mathbf{A}$ must agree with the number of rows of $\mathbf{B}$. The resultant matrix has the same number of rows as $\mathbf{A}$, and the same number of columns as $\mathbf{B}$.

$$(m \times p)(p \times n) = (m \times n).$$

2. Matrix multiplication is not commutative,

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \neq \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

3. The order of the product matter.
 - i. If we multiply $\mathbf{A}$ to the left of $\mathbf{B}$, we are pre-multiplying $\mathbf{A}$ to $\mathbf{B}$.
 - ii. If we multiply $\mathbf{A}$ to the right of $\mathbf{B}$, we are post-multiplying $\mathbf{A}$ to $\mathbf{B}$.

Properties of Matrix Multiplication

Theorem

- (i) *(Associative)* $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$.
- (ii) *(Left distributive law)* $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$.
- (iii) *(Right distributive law)* $(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$.
- (iv) *(Commute with scalar multiplication)* $c(\mathbf{AB}) = (c\mathbf{A})\mathbf{B} = \mathbf{A}(c\mathbf{B})$.
- (v) *(Multiplicative identity)* For any $m \times n$ matrix $\mathbf{A}$, $\mathbf{I}_m\mathbf{A} = \mathbf{A} = \mathbf{A}\mathbf{I}_n$.
- (vi) *(Nonzero Zero divisor)* There exists $\mathbf{A} \neq \mathbf{0}_{m \times p}$ and $\mathbf{B} \neq \mathbf{0}_{p \times n}$ such that $\mathbf{AB} = \mathbf{0}_{m \times n}$.
- (vii) *(Zero matrix)* For any $m \times n$ matrix $\mathbf{A}$, $\mathbf{A}\mathbf{0}_{n \times p} = \mathbf{0}_{m \times p}$ and $\mathbf{0}_{p \times m}\mathbf{A} = \mathbf{0}_{p \times n}$.

Remark

By the associativity of matrix multiplication, if $\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_k$ are k matrices of the right sizes such that their product is well-defined, we write it as

$$\mathbf{A}_1 \mathbf{A}_2 \cdots \mathbf{A}_k,$$

without the parenthesis (brackets).

Zero Divisors

Caution! Consider

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then

$$\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Hence, it is **false** to conclude that if $\mathbf{A}$ and $\mathbf{B}$ are matrices such that

$$\mathbf{AB} = \mathbf{0},$$

then $\mathbf{A} = \mathbf{0}$ or $\mathbf{B} = \mathbf{0}$.

Powers of Square Matrices

Definition

Define the power of **square matrices** inductively as such.

- (i) $\mathbf{A}^0 = \mathbf{I}$,
- (ii) $\mathbf{A}^n = \mathbf{A}\mathbf{A}^{n-1}$, for $n \geq 1$.

That is, $\mathbf{A}^n$ is $\mathbf{A}$ multiplied to itself n times, for $n \geq 2$. It follows that $\mathbf{A}^n\mathbf{A}^m = \mathbf{A}^{n+m}$ for positive integers m, n .

Example

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}^3 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$$

Note that powers of a matrix can only be defined for square matrices.

Question

Is it true that for any **square matrices** **A** and **B** of order n , $(\mathbf{A} + \mathbf{B})^2 = \mathbf{A}^2 + 2\mathbf{AB} + \mathbf{B}^2$?

The left hand side:

$$(\mathbf{A} + \mathbf{B})^2 = (\mathbf{A} + \mathbf{B})(\mathbf{A} + \mathbf{B}) = (\mathbf{A} + \mathbf{B})\mathbf{A} + (\mathbf{A} + \mathbf{B})\mathbf{B} = \mathbf{A}^2 + \mathbf{BA} + \mathbf{AB} + \mathbf{B}^2$$

where we in the second equality we treat $(\mathbf{A} + \mathbf{B})$ as a matrix, and use the distributive property, and used the distributive property again in the third equality.

Comparing $\mathbf{A}^2 + \mathbf{BA} + \mathbf{AB} + \mathbf{B}^2$ with the right hand side, they are equal only if

$$\mathbf{BA} + \mathbf{AB} = 2\mathbf{AB} = \mathbf{AB} + \mathbf{AB},$$

that is

$$\mathbf{BA} = \mathbf{AB},$$

which is false in general.

Transpose

Let $\mathbf{A} = (a_{ij})$ be a $m \times n$ matrix. The transpose of $\mathbf{A}$, denoted as $\mathbf{A}^T$, is the $n \times m$ matrix whose (i, j) -entry is the (j, i) -entry of $\mathbf{A}$, $\mathbf{A}^T = (b_{ij})_{n \times m}$, $b_{ij} = a_{ji}$.

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}_{m \times n} \quad \mathbf{A}^T = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}_{n \times m}$$

Examples

$$1. \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 5 \end{pmatrix}^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 5 \end{pmatrix}$$

$$2. \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}^T = (1 \quad 1 \quad 0)$$

$$3. \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix}^T = \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix}$$

Properties of Transpose

Theorem

- (i) $(\mathbf{A}^T)^T = \mathbf{A}$.
- (ii) $(c\mathbf{A})^T = c\mathbf{A}^T$.
- (iii) $(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$.
- (iv) $(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$.

The transpose provides an alternative definition of symmetric matrix. A square matrix $\mathbf{A}$ is symmetric if $\mathbf{A} = \mathbf{A}^T$.

Example

Here is an example to demonstrate the last property $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$ of the previous theorem.

$$\begin{aligned} \left(\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 3 \\ -1 & -2 \end{pmatrix} \right)^T &= \begin{pmatrix} 2 & 1 \\ 8 & 7 \end{pmatrix}^T = \begin{pmatrix} 2 & 8 \\ 1 & 7 \end{pmatrix} = \begin{pmatrix} 1 & 2 & -1 \\ 1 & 3 & -2 \end{pmatrix} \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 1 \\ 2 & 3 \\ -1 & -2 \end{pmatrix}^T \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^T \end{aligned}$$

In fact, if $\mathbf{A}$ is a $m \times p$ matrix and $\mathbf{B}$ a $p \times n$ matrix, then $\mathbf{A}^T$ is $p \times m$ and $\mathbf{B}^T$ is $n \times p$, and thus the product $\mathbf{A}^T \mathbf{B}^T$ is not well defined if $m \neq n$.

Question

Is it true that if $\mathbf{A}$ and $\mathbf{B}$ are **symmetric matrices** of the same size, then so is $\mathbf{A} + \mathbf{B}$?

Challenge

Is it true that if $\mathbf{A}$ and $\mathbf{B}$ are **symmetric matrices** of the same order, then so is $\mathbf{AB}$?

Linear System and Matrix Equation

Matrix Equation

A linear system in standard form

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

Can be expressed as a matrix equation

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}, \quad \mathbf{Ax} = \mathbf{b}$$

Here $\mathbf{A} = (a_{ij})_{m \times n}$ is called the coefficient matrix, $\mathbf{x} = (x_i)_{n \times 1}$ the variable vector, and $\mathbf{b} = (b_i)_{m \times 1}$ the constant vector.

Vector Equation

The linear system

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

can also be expressed as a vector equation:

$$x_1 \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix} + x_2 \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix} + \cdots + x_n \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}, \quad x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n = \mathbf{b}$$

Here $\mathbf{a}_i$ is called the coefficient vector for variable x_i , for $i = 1, \dots, n$.

Example

The linear system

$$\begin{cases} 3x + 2y - z = 1 \\ x + 2y + z = 3 \\ x \quad \quad + z = 2 \end{cases}$$

can be written as a matrix equation

$$\begin{pmatrix} 3 & 2 & -1 \\ 1 & 2 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$$

or a vector equation

$$x \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + y \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} + z \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}.$$

Example

The (unique) solution to the system is $x = \frac{1}{2}$, $y = \frac{1}{2}$, $z = \frac{3}{2}$. We can express the solution as a 3 by 1 matrix, $\begin{pmatrix} 1/2 \\ 1/2 \\ 3/2 \end{pmatrix}$. Check that

$$\begin{pmatrix} 3 & 2 & -1 \\ 1 & 2 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1/2 \\ 1/2 \\ 3/2 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$$

and

$$\frac{1}{2} \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} + \frac{3}{2} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}.$$

Properties of Homogeneous Linear System

Recall that a linear system is homogeneous if it has the following corresponding matrix equation

$$\mathbf{Ax} = \mathbf{0},$$

for some $m \times n$ matrix $\mathbf{A}$, a variable n -vector $\mathbf{x}$, and the $m \times 1$ zero matrix (or zero m -vector) $\mathbf{0} = \mathbf{0}_{m \times 1}$.

Theorem

A homogeneous linear system is always consistent.

Proof.

Write the homogeneous linear system as $\mathbf{Ax} = \mathbf{0}$ for some $m \times n$ matrix $\mathbf{A}$. Then

$$\mathbf{A0} = \mathbf{0},$$

that is, the $m \times 1$ zero matrix is a solution to the system. □

Properties of Homogeneous Linear System

Definition

The zero solution is called the trivial solution. If $\mathbf{x} \neq \mathbf{0}$ is a nonzero solution to the homogeneous system, it is called a nontrivial solution.

Theorem

A homogeneous linear system has *infinitely many solutions* if and only if it has a *nontrivial solution*.

Proof.

($\Rightarrow$) If the system has infinitely many solutions, it must surely have a nontrivial solution.

($\Leftarrow$) Suppose now $\mathbf{u} \neq \mathbf{0}$ is a nontrivial solution to the homogeneous system $\mathbf{Ax} = \mathbf{0}$, that is, $\mathbf{Au} = \mathbf{0}$. Now for any real number $s \in \mathbb{R}$, since scalar multiplication commutes with matrix multiplication,

$$\mathbf{A}(s\mathbf{u}) = s(\mathbf{Au}) = s\mathbf{0} = \mathbf{0},$$

that is, $s\mathbf{u}$ is a solution to the homogeneous linear system too. Hence, the system has infinitely many solutions. □

Example

Consider the following homogeneous linear system

$$\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

See that $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is a nontrivial solution to the system,

$$\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Now for any real number $s \in \mathbb{R}$, $s \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} s \\ 0 \end{pmatrix}$ is a solution to the system too,

$$\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} s \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Question

Show that if the **trivial solution** is a solution to the linear system, then it must be a homogeneous linear system.

Question

Which of the following statements is/are true?

1. If the homogeneous system has a **unique solution**, it must be the **trivial solution**.
2. If the homogeneous system has the **trivial solution**, it must be the **unique solution**.

Solutions to Homogeneous and Non-homogeneous Linear System

Lemma

Let $\mathbf{v}$ be a particular solution $\mathbf{Ax} = \mathbf{b}$, and $\mathbf{u}$ be a particular solution to the *homogeneous system* $\mathbf{Ax} = \mathbf{0}$ with the same coefficient matrix $\mathbf{A}$. Then $\mathbf{v} + \mathbf{u}$ is also a solution to $\mathbf{Ax} = \mathbf{b}$.

Proof.

By hypothesis, $\mathbf{Av} = \mathbf{b}$ and $\mathbf{Au} = \mathbf{0}$. Hence, by the distribution properties of matrix multiplication,

$$\mathbf{A}(\mathbf{v} + \mathbf{u}) = \mathbf{Av} + \mathbf{Au} = \mathbf{b} + \mathbf{0} = \mathbf{b}.$$



Lemma

Suppose $\mathbf{v}_1$ and $\mathbf{v}_2$ are solutions to the linear system $\mathbf{Ax} = \mathbf{b}$. Then $\mathbf{v}_1 - \mathbf{v}_2$ is a solution to the homogeneous linear system $\mathbf{Ax} = \mathbf{0}$ with the same coefficient matrix.

Proof.

By hypothesis, $\mathbf{Av}_1 = \mathbf{b}$, $\mathbf{Av}_2 = \mathbf{b}$. Hence,

$$\mathbf{A}(\mathbf{v}_1 - \mathbf{v}_2) = \mathbf{Av}_1 - \mathbf{Av}_2 = \mathbf{b} - \mathbf{b} = \mathbf{0}.$$

Challenge

Let $\mathbf{v}$ be a particular solution to a non-homogeneous system $\mathbf{Ax} = \mathbf{b}$, $\mathbf{b} \neq \mathbf{0}$.

Show that

$$\mathbf{v} + s_1\mathbf{u}_1 + s_2\mathbf{u}_2 + \cdots + s_k\mathbf{u}_k, \quad s_1, s_2, \dots, s_k \in \mathbb{R}$$

is the general solution to $\mathbf{Ax} = \mathbf{b}$ if and only if

$$s_1\mathbf{u}_1 + s_2\mathbf{u}_2 + \cdots + s_k\mathbf{u}_k, \quad s_1, s_2, \dots, s_k \in \mathbb{R}$$

is the general solution to the homogeneous linear system $\mathbf{Ax} = \mathbf{0}$.

Introduction to Submatrices

Let $\mathbf{A}$ be an $m \times n$ matrix,

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

Rows of $\mathbf{A}$:

$$\begin{aligned} \mathbf{r}_1 &= (a_{11} \quad a_{12} \quad \cdots \quad a_{1n}) \\ \mathbf{r}_2 &= (a_{21} \quad a_{22} \quad \cdots \quad a_{2n}) \\ &\vdots \\ \mathbf{r}_m &= (a_{m1} \quad a_{m2} \quad \cdots \quad a_{mn}) \end{aligned}$$

Columns of $\mathbf{A}$:

$$\mathbf{c}_1 = \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix} \quad \mathbf{c}_2 = \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix} \quad \cdots \quad \mathbf{c}_n = \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix}$$

The rows and columns of a matrix are examples of submatrices of a matrix.

Submatrix

Definition

A $p \times q$ submatrix of an $m \times n$ matrix $\mathbf{A}$, $p \leq m$, $q \leq n$, is formed by taking a $p \times q$ block of the entries of the matrix $\mathbf{A}$.

Example

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 5 & 7 & 1 \\ 2 & 4 & 6 & 1 & 2 \\ 1 & 2 & 2 & 1 & 3 \end{pmatrix}.$$

- (i) Each row is a 1 by 5 submatrix, each column is a 3 by 1 submatrix.
- (ii) $\begin{pmatrix} 2 & 4 & 6 \\ 1 & 2 & 2 \end{pmatrix}$ is a 2 by 3 submatrix obtained by taking rows 2 and 3, and columns 1 to 3.
- (iii) $\begin{pmatrix} 5 & 7 \\ 6 & 1 \\ 2 & 1 \end{pmatrix}$ is a 3 by 2 submatrix obtained by taking rows 1 to 3, and columns 3 and 4.

Block Multiplication

Theorem

Let $\mathbf{A}$ be an $m \times p$ matrix and $\mathbf{B}$ a $p \times n$ matrix. Let $\mathbf{A}_1$ be a $(m_2 - m_1 + 1) \times p$ submatrix of $\mathbf{A}$ obtained by taking rows m_1 to m_2 , and $\mathbf{b}_1$ a $p \times (n_2 - n_1 + 1)$ submatrix of $\mathbf{B}$ obtained by taking columns n_1 to n_2 . Then the product $\mathbf{A}_1 \mathbf{b}_1$ is a $(m_2 - m_1 + 1) \times (n_2 - n_1 + 1)$ submatrix of $\mathbf{AB}$ obtained by taking rows m_1 to m_2 and columns n_1 to n_2 .

We call this block multiplication.

$$\begin{array}{c}
 \begin{pmatrix} m_1 \\ \vdots \\ m_2 \end{pmatrix} \left(\begin{array}{ccc} & & \\ & \mathbf{A}_1 & \\ & & \end{array} \right) \begin{pmatrix} b_{11} & \dots & b_{1n_2} & \dots & b_{1n} \\ b_{21} & \dots & b_{2n_2} & \dots & b_{2n} \\ \vdots & \dots & \vdots & \dots & \vdots \\ b_{p1} & \dots & b_{pn_2} & \dots & b_{pn} \end{pmatrix} = \begin{pmatrix} \text{g} & \text{g} & \text{g} \\ \vdots & \vdots & \vdots \\ \text{g} & \text{g} & \text{g} \end{pmatrix} \begin{pmatrix} n_1 & n_2 & \dots & n \end{pmatrix} \\
 \mathbf{A} \qquad \mathbf{B} \qquad \mathbf{AB}
 \end{array}$$

The diagram illustrates the block multiplication process. On the left, matrix $\mathbf{A}$ is shown with a pink block $\mathbf{A}_1$ spanning rows m_1 to m_2 . Matrix $\mathbf{B}$ is shown with a pink block $\mathbf{B}_1$ spanning columns n_1 to n_2 . The product $\mathbf{AB}$ is shown on the right, with the resulting block $\mathbf{A}_1 \mathbf{B}_1$ highlighted in pink, spanning rows m_1 to m_2 and columns n_1 to n_2 . The matrices are represented as large parentheses containing their respective elements or blocks.

Example

Let $\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 2 \\ 3 & 2 & 1 \\ 4 & 4 & 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 3 & 5 & 7 & 1 \\ 2 & 4 & 6 & 1 & 2 \\ 1 & 2 & 2 & 1 & 3 \end{pmatrix}$. Then

$$\mathbf{AB} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 2 \\ 3 & 2 & 1 \\ 4 & 4 & 2 \end{pmatrix} \begin{pmatrix} 1 & 3 & 5 & 7 & 1 \\ 2 & 4 & 6 & 1 & 2 \\ 1 & 2 & 2 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 5 & 7 & 8 & 4 \\ 7 & 15 & 21 & 11 & 11 \\ 8 & 19 & 29 & 24 & 10 \\ 14 & 32 & 48 & 34 & 18 \end{pmatrix}.$$

(i) $\mathbf{A}_1 = \begin{pmatrix} 1 & 2 & 2 \\ 3 & 2 & 1 \end{pmatrix}$ is obtained from $\mathbf{A}$ by taking rows 2 and 3, and $\mathbf{B}_1 = \begin{pmatrix} 5 & 7 \\ 6 & 1 \\ 2 & 1 \end{pmatrix}$ is obtained from $\mathbf{B}$ by taking

columns 3 and 4. Then $\mathbf{A}_1\mathbf{B}_1 = \begin{pmatrix} 21 & 11 \\ 29 & 24 \end{pmatrix}$ is a submatrix of $\mathbf{AB}$ by taking rows 2 and 3, columns 3 and 4.

(ii) $\mathbf{A}_1 = \begin{pmatrix} 3 & 2 & 1 \\ 4 & 4 & 2 \end{pmatrix}$ is obtained from $\mathbf{A}$ by taking rows 3 and 4, and $\mathbf{B}_1 = \begin{pmatrix} 3 \\ 4 \\ 2 \end{pmatrix}$ is column 2 of $\mathbf{B}$. Then

$\mathbf{A}_1\mathbf{B}_1 = \begin{pmatrix} 19 \\ 32 \end{pmatrix}$ is a submatrix of $\mathbf{AB}$ by taking rows 3 and 4, and column 2.

Question

Given that

$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 2 \\ 3 & 2 & 1 \\ 4 & 4 & 2 \end{pmatrix} \begin{pmatrix} 1 & 3 & 5 & 7 & 1 \\ 2 & 4 & 6 & 1 & 2 \\ 1 & 2 & 2 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 5 & 7 & 8 & 4 \\ 7 & 15 & 21 & 11 & 11 \\ 8 & 19 & 29 & 24 & 10 \\ 14 & 32 & 48 & 34 & 18 \end{pmatrix}.$$

1. What is $\begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 2 \\ 3 & 2 & 1 \\ 4 & 4 & 2 \end{pmatrix} \begin{pmatrix} 3 & 5 \\ 4 & 6 \\ 2 & 2 \end{pmatrix}$?

2. What is $(4 \ 4 \ 2) \begin{pmatrix} 1 & 3 & 5 & 7 & 1 \\ 2 & 4 & 6 & 1 & 2 \\ 1 & 2 & 2 & 1 & 3 \end{pmatrix}$?

Block Multiplication

In particular, let $\mathbf{b}_j$ be the j -th column of $\mathbf{B}$. Then

$$\mathbf{AB} = \mathbf{A} \begin{pmatrix} \left| \begin{smallmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \vdots \\ \mathbf{b}_n \end{smallmatrix} \right| \end{pmatrix} = \begin{pmatrix} \left| \begin{smallmatrix} \mathbf{Ab}_1 \\ \mathbf{Ab}_2 \\ \vdots \\ \mathbf{Ab}_n \end{smallmatrix} \right| \end{pmatrix}.$$

That is, the j -th column of the product $\mathbf{AB}$ is the product of $\mathbf{A}$ with the j -th column of $\mathbf{B}$.

Also, if $\mathbf{a}_i$ is the i -th row of $\mathbf{A}$, then

$$\mathbf{AB} = \begin{pmatrix} - & \mathbf{a}_1 & - \\ - & \mathbf{a}_2 & - \\ & \vdots & \\ - & \mathbf{a}_m & - \end{pmatrix} \mathbf{B} = \begin{pmatrix} - & \mathbf{a}_1 \mathbf{B} & - \\ - & \mathbf{a}_2 \mathbf{B} & - \\ & \vdots & \\ - & \mathbf{a}_m \mathbf{B} & - \end{pmatrix}.$$

That is, the i -th row of the product $\mathbf{AB}$ is the product of the i -th row of $\mathbf{A}$ with $\mathbf{B}$.

Solving Matrix Equations

Let $\mathbf{A} = \begin{pmatrix} 3 & 2 & -1 \\ 5 & -1 & 3 \\ 2 & 1 & -1 \end{pmatrix}$. Find a 3×3 matrix $\mathbf{X}$ such that

$$\mathbf{AX} = \begin{pmatrix} 3 & 2 & -1 \\ 5 & -1 & 3 \\ 2 & 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 3 & 1 & 0 \end{pmatrix}.$$

By block multiplication, we are solving for the 3 linear systems

$$\mathbf{A} \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{A} \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{A} \begin{pmatrix} x_3 \\ y_3 \\ z_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

Example

Solve the first linear system,

$$\begin{pmatrix} 3 & 2 & -1 & | & 1 \\ 5 & -1 & 3 & | & 2 \\ 2 & 1 & -1 & | & 3 \end{pmatrix} \xrightarrow[R_3 - \frac{2}{3}R_1]{R_2 - \frac{5}{3}R_1} \begin{pmatrix} 3 & 2 & -1 & | & 1 \\ 0 & -13/3 & 14/3 & | & 1/3 \\ 0 & -1/3 & -1/3 & | & 7/3 \end{pmatrix} \xrightarrow[-\frac{3}{13}R_2]{R_3 - \frac{1}{13}R_2} \begin{pmatrix} 3 & 2 & -1 & | & 1 \\ 0 & 1 & -14/13 & | & -1/13 \\ 0 & 0 & 1 & | & -10/3 \end{pmatrix}$$
$$\xrightarrow[R_1 + R_3]{R_2 + \frac{14}{13}R_3} \begin{pmatrix} 3 & 2 & 0 & | & -7/3 \\ 0 & 1 & 0 & | & -11/3 \\ 0 & 0 & 1 & | & -10/3 \end{pmatrix} \xrightarrow{\frac{1}{3}R_1}{R_1 - 2R_2} \begin{pmatrix} 1 & 0 & 0 & | & 5/3 \\ 0 & 1 & 0 & | & -11/3 \\ 0 & 0 & 1 & | & -10/3 \end{pmatrix}$$

$$\text{So, } \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 5 \\ -11 \\ 10 \end{pmatrix}.$$

Example

Solve the second linear system,

$$\begin{pmatrix} 3 & 2 & -1 & \big| & 2 \\ 5 & -1 & 3 & \big| & 1 \\ 2 & 1 & -1 & \big| & 1 \end{pmatrix} \xrightarrow[R_3 - \frac{2}{3}R_1]{R_2 - \frac{5}{3}R_1} \begin{pmatrix} 3 & 2 & -1 & \big| & 2 \\ 0 & -13/3 & 14/3 & \big| & -7/3 \\ 0 & -1/3 & -1/3 & \big| & -1/3 \end{pmatrix} \xrightarrow[R_2 - \frac{1}{13}R_1]{R_3 - \frac{1}{13}R_2} \begin{pmatrix} 3 & 2 & -1 & \big| & 2 \\ 0 & 1 & -14/13 & \big| & 7/13 \\ 0 & 0 & 1 & \big| & 2/9 \end{pmatrix} \\ \xrightarrow[R_1 + R_3]{R_2 + \frac{14}{13}R_3} \begin{pmatrix} 3 & 2 & 0 & \big| & 20/9 \\ 0 & 1 & 0 & \big| & 7/9 \\ 0 & 0 & 1 & \big| & 2/9 \end{pmatrix} \xrightarrow{\frac{1}{3}R_1} \begin{pmatrix} 1 & 0 & 0 & \big| & 2/9 \\ 0 & 1 & 0 & \big| & 7/9 \\ 0 & 0 & 1 & \big| & 2/9 \end{pmatrix}$$

So, $\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} = \frac{1}{9} \begin{pmatrix} 2 \\ 7 \\ 2 \end{pmatrix}$. The row operations used here in the reduction process is exactly those in solving the first linear system. This is because they both have the same coefficient matrix.

We can conclude likewise that the exact same row operations could be used to solve the third linear system. Hence, we could instead solve all 3 linear systems simultaneously!

Combining Augmented Matrices

Solve the following 3 linear systems

$$\begin{array}{rrcr} 3x & + & 2y & - & z & = & a \\ 5x & - & y & + & 3z & = & b \\ 2x & + & y & - & z & = & c \end{array}, \text{ for } \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

Matrix equation:

$$\begin{pmatrix} 3 & 2 & -1 \\ 5 & -1 & 3 \\ 2 & 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 3 & 1 & 0 \end{pmatrix}$$

Augmented matrix:

$$\left(\begin{array}{ccc|c|c|c} 3 & 2 & -1 & 1 & 2 & 1 \\ 5 & -1 & 3 & 2 & 1 & 1 \\ 2 & 1 & -1 & 3 & 1 & 0 \end{array} \right)$$

Solve the 3 linear system simultaneously.

Example

$$\begin{aligned} & \left(\begin{array}{ccc|ccc} 3 & 2 & -1 & 1 & 2 & 1 \\ 5 & -1 & 3 & 2 & 1 & 1 \\ 2 & 1 & -1 & 3 & 1 & 0 \end{array} \right) \xrightarrow[R_3 - \frac{2}{3}R_1]{R_2 - \frac{5}{3}R_1} \left(\begin{array}{ccc|ccc} 3 & 2 & -1 & 1 & 2 & 1 \\ 0 & -13/3 & 14/3 & 1/3 & -7/3 & -2/3 \\ 0 & -1/3 & -1/3 & 7/3 & -1/3 & -2/3 \end{array} \right) \xrightarrow[R_3 - \frac{1}{13}R_2]{-\frac{13}{9}R_3} \\ & \left(\begin{array}{ccc|ccc} 3 & 2 & -1 & 1 & 2 & 1 \\ 0 & 1 & -14/13 & -1/13 & 7/13 & 2/13 \\ 0 & 0 & 1 & -10/3 & 2/9 & 8/9 \end{array} \right) \xrightarrow[R_1 + R_3]{R_2 + \frac{14}{13}R_3} \left(\begin{array}{ccc|ccc} 3 & 2 & 0 & -7/3 & 20/9 & 17/9 \\ 0 & 1 & 0 & -11/3 & 7/9 & 10/9 \\ 0 & 0 & 1 & -10/3 & 2/9 & 8/9 \end{array} \right) \\ & \xrightarrow[R_1 - 2R_2]{\frac{1}{3}R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 5/3 & 2/9 & -1/9 \\ 0 & 1 & 0 & -11/3 & 7/9 & 10/9 \\ 0 & 0 & 1 & -10/3 & 2/9 & 8/9 \end{array} \right) \\ & \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 5 \\ -11 \\ -10 \end{pmatrix}, \quad \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} = \frac{1}{9} \begin{pmatrix} 2 \\ 7 \\ 2 \end{pmatrix}, \quad \begin{pmatrix} x_3 \\ y_3 \\ z_3 \end{pmatrix} = \frac{1}{9} \begin{pmatrix} -1 \\ 10 \\ 8 \end{pmatrix}. \end{aligned}$$

That is,

$$\mathbf{x} = \begin{pmatrix} 5/3 & 2/9 & -1/9 \\ -11/3 & 7/9 & 10/9 \\ -10/3 & 2/9 & 8/9 \end{pmatrix}$$

Combining Augmented Matrices

In general: p linear systems with the same coefficient matrix $\mathbf{A} = (a_{ij})_{m \times n}$, for $k = 1, \dots, p$, is equivalent to solving a matrix equation

$$\left\{ \begin{array}{l} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_{1k} \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_{2k} \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_{mk} \end{array} \right.$$

$$\Leftrightarrow \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix} = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mp} \end{pmatrix}$$

$$\left(\begin{array}{cccc|cc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_{11} & b_{12} & \vdots \\ a_{21} & a_{22} & \cdots & a_{2n} & b_{21} & b_{22} & \vdots \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_{m1} & b_{m2} & b_{mp} \end{array} \right)$$

Inverse of Matrices

Introduction

1. Suppose $ab = ac$ for some $a, b, c \in \mathbb{R}$ with $a \neq 0$, then $b = c$.
2. This is because if $a \neq 0$, $\frac{1}{a}$ exists, and $a\frac{1}{a} = 1$.
3. Hence, multiplying both sides of $ab = ac$ with $\frac{1}{a}$, we have $b = 1 \times b = \frac{1}{a}ab = \frac{1}{a}ac = 1 \times c = c$.
4. This is used to solve $ax = b$ for some $a, b \in \mathbb{R}$. We can conclude that $x = \frac{b}{a}$.
5. Ideally, we want to apply the same idea to solve a linear system,

$$\mathbf{Ax} = \mathbf{b} \quad \Rightarrow \quad \mathbf{x} = \frac{\mathbf{b}}{\mathbf{A}}.$$

6. Idea: Try to define $\frac{1}{\mathbf{A}}$.

Introduction

1. To define $\frac{1}{\mathbf{A}}$, we need the matrix equivalent of 1, since $\frac{1}{a}$ is defined such that $\frac{1}{a}a = 1$.
2. 1 is the multiplicative identity of the real numbers, $1 \times a = a$ for any $a \in \mathbb{R}$.
3. The matrix multiplicative identity is the identity matrix, $\mathbf{I}_n$,

$$\mathbf{I}\mathbf{A} = \mathbf{A} = \mathbf{A}\mathbf{I}$$

for any matrix $\mathbf{A}$.

4. So, define $\frac{1}{\mathbf{A}}$ as the matrix such that $\frac{1}{\mathbf{A}}\mathbf{A} = \mathbf{I}$.
5. But because matrix multiplication is not commutative, we need a matrix $\mathbf{B}$ such that

$$\mathbf{A}\mathbf{B} = \mathbf{I} \quad \text{and} \quad \mathbf{B}\mathbf{A} = \mathbf{I}$$

Inverse of Square Matrices

Definition

A $n \times n$ square matrix $\mathbf{A}$ is invertible if there exists a matrix $\mathbf{B}$ such that

$$\mathbf{AB} = \mathbf{I}_n = \mathbf{BA}.$$

A matrix is said to be non-invertible otherwise.

Example

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

Remarks

1. By definition, for a **square matrix** $\mathbf{A}$ to be invertible, there must be a $\mathbf{B}$ such that $\mathbf{BA} = \mathbf{I}_n$ **AND** $\mathbf{AB} = \mathbf{I}_n$ simultaneously.
2. Turns out that $\mathbf{A}$ is invertible
 - i. $\Leftrightarrow$ there is a $\mathbf{B}$ such that $\mathbf{AB} = \mathbf{I}$ ($\mathbf{A}$ has a right inverse),
 - ii. $\Leftrightarrow$ there is a $\mathbf{B}$ such that $\mathbf{BA} = \mathbf{I}$ ($\mathbf{A}$ has a left inverse).
3. Moreover, $\mathbf{BA} = \mathbf{I}_n \Leftrightarrow \mathbf{AB} = \mathbf{I}_n$ (left inverse = right inverse).
4. All **non-square** matrices are **not invertible**. If $\mathbf{A}$ is a $n \times m$ matrix with $n \neq m$, then for $\mathbf{AB} = \mathbf{I}_n$, $\mathbf{B}$ must be of size $m \times n$, and hence $\mathbf{BA} \neq \mathbf{I}_n$.

Uniqueness of inverse

Theorem

If $\mathbf{B}$ and $\mathbf{C}$ are both inverses of a square matrix $\mathbf{A}$, then $\mathbf{B} = \mathbf{C}$.

Proof.

Suppose $\mathbf{B}$ and $\mathbf{C}$ are such that $\mathbf{AB} = \mathbf{I} = \mathbf{BA}$ and $\mathbf{AC} = \mathbf{I} = \mathbf{CA}$. Then by associativity of matrix multiplication,

$$\mathbf{B} = \mathbf{BI} = \mathbf{B(AC)} = (\mathbf{BA})\mathbf{C} = \mathbf{IC} = \mathbf{C}.$$



So since the inverse is unique, we can denote the inverse of an invertible matrix $\mathbf{A}$ by $\mathbf{A}^{-1}$ and call it the inverse of $\mathbf{A}$.

Question: Is the identity matrix $\mathbf{I}$ invertible? If it is, what is its inverse?

Non-Invertible Square Matrix

Not all matrices are invertible. For example, consider the matrix $\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$.

It is not invertible since for any order 2 square matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ a+c & b+d \end{pmatrix} \neq \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Definition

A **non-invertible** square matrix is called a singular matrix.

Remark

Some textbook use the term singular interchangeably with non-invertible, that is, they do not insist that singular matrices are square matrices.

Inverse of 2 by 2 Square Matrices

Theorem

A 2×2 *square* matrix $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is invertible if and only if $ad - bc \neq 0$. In this case, the *inverse* is given by

$$\mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Cancellation Law for Matrices

Theorem

Let **A** be an *invertible* matrix of order n .

- (i) (Left cancellation) If **B** and **C** are $n \times m$ matrices with $\mathbf{AB} = \mathbf{AC}$, then $\mathbf{B} = \mathbf{C}$.
- (ii) (Right cancellation) If **B** and **C** are $m \times n$ matrices with $\mathbf{BA} = \mathbf{CA}$, then $\mathbf{B} = \mathbf{C}$.

Proof.

(i)

$$\mathbf{AB} = \mathbf{AC} \Rightarrow \mathbf{B} = \mathbf{A}^{-1}\mathbf{AB} = \mathbf{A}^{-1}\mathbf{AC} = \mathbf{C}.$$

(ii)

$$\mathbf{BA} = \mathbf{CA} \Rightarrow \mathbf{B} = \mathbf{BAA}^{-1} = \mathbf{CAA}^{-1} = \mathbf{C}.$$



Caution

If $\mathbf{AB} = \mathbf{CA}$, we cannot conclude that $\mathbf{B} = \mathbf{C}$.

Question

Suppose $\mathbf{A}$ is invertible. Can we conclude that the system $\mathbf{Ax} = \mathbf{b}$ is consistent? If so, what is a solution, and is it unique?

Invertibility and Linear System

Theorem

Suppose $\mathbf{A}$ is an $n \times n$ invertible *square* matrix. Then for any $n \times 1$ vector $\mathbf{b}$, $\mathbf{Ax} = \mathbf{b}$ has a unique solution.

Proof.

There are two claims in the theorem; it claims that the system $\mathbf{Ax} = \mathbf{b}$ is consistent, and that the solution is unique. Firstly, we will check that $\mathbf{u} = \mathbf{A}^{-1}\mathbf{b}$ is a solution. Indeed,

$$\mathbf{Au} = \mathbf{A}(\mathbf{A}^{-1}\mathbf{b}) = \mathbf{b}.$$

Now suppose $\mathbf{v}$ is also a solution, that is, $\mathbf{Av} = \mathbf{b}$. Then

$$\mathbf{Av} = \mathbf{b} = \mathbf{Au}.$$

By the cancellation law, we have $\mathbf{v} = \mathbf{u} = \mathbf{A}^{-1}\mathbf{b}$. Hence, $\mathbf{A}^{-1}\mathbf{b}$ is the unique solution to the system. $\square$

Invertibility and Linear System

Corollary

Suppose $\mathbf{A}$ is *invertible*. Then the *trivial solution* is the *only solution* to the homogeneous system $\mathbf{Ax} = \mathbf{0}$.

Proof.

This follows immediately from the previous theorem by letting $\mathbf{b} = \mathbf{0}$. □

Example

Consider that linear system

$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}.$$

Let $\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$. Readers should check that $\mathbf{A}^{-1} = \begin{pmatrix} 0.5 & 0.5 & -0.5 \\ -0.5 & 0.5 & 0.5 \\ 0.5 & -0.5 & 0.5 \end{pmatrix}$. Solving the system, we have

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 4 \\ 0 & 1 & 1 & 6 \end{array} \right) \xrightarrow{RREF} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 2 \end{array} \right).$$

The unique solution is $\begin{pmatrix} 0 \\ 4 \\ 2 \end{pmatrix}$, which is exactly equal to $\mathbf{A}^{-1}\mathbf{b}$,

$$\mathbf{A}^{-1} \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} = \begin{pmatrix} 0.5 & 0.5 & -0.5 \\ -0.5 & 0.5 & 0.5 \\ 0.5 & -0.5 & 0.5 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \\ 2 \end{pmatrix}.$$

Exercise

Verify that the homogeneous system

$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

has only the trivial solution.

Algorithm to Computing Inverse

Suppose $\mathbf{A}$ is an invertible $n \times n$ matrix. By uniqueness of the inverse, there must be a unique solution to

$$\mathbf{AX} = \mathbf{I}.$$

By block multiplication, we are solving the augmented matrix

$$(\mathbf{A} \mid \mathbf{I}) \xrightarrow{RREF} (\mathbf{I} \mid \mathbf{A}^{-1}).$$

This also gives us an algorithm to check if a square matrix is invertible. For if we reduce the combined augmented matrix

$$(\mathbf{A} \mid \mathbf{I}_n)$$

and the coefficient side is not the identity matrix in the RREF, we can conclude that $\mathbf{A}$ is not invertible.

Example

1. Find the inverse of $\begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$.

$$\left(\begin{array}{cc|cc} 3 & 5 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array} \right) \xrightarrow{R_1 - 3R_2} \xrightarrow{R_2 + 2R_1} \left(\begin{array}{cc|cc} 0 & -1 & 1 & -3 \\ 1 & 0 & 2 & -5 \end{array} \right) \xrightarrow{-R_1} \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|cc} 1 & 0 & 2 & -5 \\ 0 & 1 & -1 & 3 \end{array} \right).$$

So, $\begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}^{-1} = \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix}$. Indeed, this is exactly the answer obtained if we used

$$\frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{(3)(2) - (5)(1)} \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix}.$$

Example

2. Find the inverse of $\begin{pmatrix} 2 & 7 & 1 \\ 1 & 4 & -1 \\ 1 & 3 & 0 \end{pmatrix}$.

$$\left(\begin{array}{ccc|ccc} 2 & 7 & 1 & 1 & 0 & 0 \\ 1 & 4 & -1 & 0 & 1 & 0 \\ 1 & 3 & 0 & 0 & 0 & 1 \end{array} \right) \xrightarrow{RREF} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -3/2 & -3/2 & 11/2 \\ 0 & 1 & 0 & 1/2 & 1/2 & -3/2 \\ 0 & 0 & 1 & 1/2 & -1/2 & -1/2 \end{array} \right).$$

$$\text{So, } \begin{pmatrix} 2 & 7 & 1 \\ 1 & 4 & -1 \\ 1 & 3 & 0 \end{pmatrix}^{-1} = \frac{1}{2} \begin{pmatrix} -3 & -3 & 11 \\ 1 & 1 & -3 \\ 1 & -1 & -1 \end{pmatrix}.$$

Properties of Inverses

Theorem

Let $\mathbf{A}$ be an *invertible matrix* of order n .

- (i) $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$.
- (ii) For any *nonzero* real number $a \in \mathbb{R}$, $(a\mathbf{A})$ is *invertible* with *inverse* $(a\mathbf{A})^{-1} = \frac{1}{a}\mathbf{A}^{-1}$.
- (iii) $\mathbf{A}^T$ is *invertible* with *inverse* $(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T$.
- (iv) If $\mathbf{B}$ is an *invertible* matrix of order n , then $(\mathbf{AB})$ is *invertible* with *inverse* $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.

Partial Proof.

The proof of (i) to (iii) is left as a exercise. For (iv),

$$(\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{AB}) = \mathbf{B}^{-1}(\mathbf{A}^{-1}\mathbf{A})\mathbf{B} = \mathbf{B}^{-1}\mathbf{B} = \mathbf{I} = \mathbf{AA}^{-1} = \mathbf{A}(\mathbf{BB}^{-1})\mathbf{A}^{-1} = (\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}).$$



Remark

1. The precise statement for (iii) is that as follows.

A square matrix $\mathbf{A}$ is invertible if and only if its transpose $\mathbf{A}^T$ is. In this case, the inverse of its transpose is the transpose of its inverse, $(\mathbf{A}^{-1})^T$.

2. **Caution:** $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} \neq \mathbf{A}^{-1}\mathbf{B}^{-1}$.

3. If $\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_k$ are invertible, the product $\mathbf{A}_1\mathbf{A}_2 \cdots \mathbf{A}_k$ is invertible with $(\mathbf{A}_1\mathbf{A}_2 \cdots \mathbf{A}_k)^{-1} = \mathbf{A}_k^{-1} \cdots \mathbf{A}_2^{-1}\mathbf{A}_1^{-1}$ if $\mathbf{A}_i$ is an invertible matrix for $i = 1, \dots, k$.

4. The negative power of an invertible matrix is defined to be

$$\mathbf{A}^{-n} = (\mathbf{A}^{-1})^n$$

for any $n > 0$.

Question

Is the **inverse** of an **invertible** symmetric matrix **symmetric**?

Elementary Matrices

Elementary Matrices

Definition

A square matrix $\mathbf{E}$ of order n is called an elementary matrix if it can be obtained from the identity matrix $\mathbf{I}_n$ by performing a single elementary row operation

$$\mathbf{I}_n \xrightarrow{r} \mathbf{E},$$

where r is an elementary row operation. The elementary row operation is said to be the row operation corresponding to the elementary matrix.

Example

$$\begin{aligned} 1. \mathbf{I}_4 \xrightarrow{R_2+3R_4} \mathbf{E} &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} & 2. \mathbf{I}_4 \xrightarrow{R_1 \leftrightarrow R_3} \mathbf{E} &= \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \\ 3. \mathbf{I}_4 \xrightarrow{3R_2} \mathbf{E} &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \end{aligned}$$

Elementary Matrices and Elementary Row Operations

Let $\mathbf{A}$ be an $n \times m$ matrix and let $\mathbf{E}$ be the $n \times n$ elementary matrix corresponding to the elementary row operation r . Then the product $\mathbf{EA}$ is the resultant of performing the row operation r on $\mathbf{A}$,

$$\mathbf{A} \xrightarrow{r} \mathbf{EA}.$$

That is, performing elementary row operations is equivalent to premultiplying by the corresponding elementary matrix.

Example

$$1. \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & -1 \\ 2 & 1 & 4 & 2 \end{pmatrix} \xrightarrow{R_2+2R_1} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 3 & 4 & 5 & -1 \\ 2 & 1 & 4 & 2 \end{pmatrix}$$
$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & -1 \\ 2 & 1 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 3 & 4 & 5 & -1 \\ 2 & 1 & 4 & 2 \end{pmatrix}$$

Example

$$2. \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & -1 \\ 2 & 1 & 4 & 2 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 4 & 2 \\ 1 & 2 & 3 & -1 \end{pmatrix}$$
$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & -1 \\ 2 & 1 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 4 & 2 \\ 1 & 2 & 3 & -1 \end{pmatrix}$$

$$3. \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & -1 \\ 2 & 1 & 4 & 2 \end{pmatrix} \xrightarrow{-2R_2} \begin{pmatrix} 1 & 1 & 1 & 0 \\ -2 & -4 & -6 & 2 \\ 2 & 1 & 4 & 2 \end{pmatrix}$$
$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & -1 \\ 2 & 1 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 0 \\ -2 & -4 & -6 & 2 \\ 2 & 1 & 4 & 2 \end{pmatrix}$$

Row Equivalent Matrices

Recall that matrices **A** and **B** are said to be **row equivalent** if **B** can be obtained from **A** by performing a series of elementary row operations.

Suppose now **B** is row equivalent to **A**,

$$\mathbf{A} \xrightarrow{r_1} \xrightarrow{r_2} \cdots \xrightarrow{r_k} \mathbf{B}.$$

Let $\mathbf{E}_i$ be the elementary matrix corresponding to the row operation r_i , for $i = 1, 2, \dots, k$. Then

$$\mathbf{B} = \mathbf{E}_k \cdots \mathbf{E}_2 \mathbf{E}_1 \mathbf{A}.$$

This follows from the previous discussion,

$$\mathbf{A} \xrightarrow{r_1} \mathbf{E}_1 \mathbf{A} \xrightarrow{r_2} \mathbf{E}_2 \mathbf{E}_1 \mathbf{A} \xrightarrow{r_3} \cdots \xrightarrow{r_k} \mathbf{E}_k \cdots \mathbf{E}_2 \mathbf{E}_1 \mathbf{A} = \mathbf{B}.$$

Example

$$\begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & -1 \\ 2 & 1 & 4 & 2 \end{pmatrix} \xrightarrow{R_2+2R_1} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 4 & 2 \\ 3 & 4 & 5 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & -1 \\ 2 & 1 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 3 & 4 & 5 & -1 \\ 2 & 1 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 4 & 2 \\ 3 & 4 & 5 & -1 \end{pmatrix}$$

Question

Consider the following

$$\mathbf{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_2+3R_1} \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \mathbf{B}$$

Give a **single** row operation that reduces $\mathbf{B}$ to the identity matrix $\mathbf{I}_3$,

$$\mathbf{B} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{???} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Question

Consider the following

$$\mathbf{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_2+3R_1} \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \mathbf{B}$$

Give a **single** row operation that reduces **B** to the identity matrix $\mathbf{I}_3$,

$$\mathbf{B} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_2-3R_1} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Inverse of Elementary Matrices

Recall that every elementary row operation has an inverse, and that performing a row operation is equivalent to premultiplying by the corresponding elementary matrix.

Consider now a row operation r , with its corresponding elementary matrix, $\mathbf{E}$. Let r' be the reverse of the row operation r , and $\mathbf{E}'$ its corresponding elementary matrix. By definition, $\mathbf{I} \xrightarrow{r} \mathbf{E}$. Now if we apply r' to $\mathbf{E}$, since it is the reverse of r , we should get back the identity matrix,

$$\mathbf{I} \xrightarrow{r} \mathbf{E} \xrightarrow{r'} \mathbf{I}.$$

Hence, we have

$$\mathbf{I} = \mathbf{E}'\mathbf{E} = \mathbf{E}'\mathbf{E}.$$

Similarly, $\mathbf{I} \xrightarrow{r'} \mathbf{E}' \xrightarrow{r} \mathbf{I}$ tells us that

$$\mathbf{I} = \mathbf{E}\mathbf{E}' = \mathbf{E}\mathbf{E}'.$$

This shows that $\mathbf{E}$ is invertible with inverse $\mathbf{E}'$, which is also an elementary matrix.

Inverse of Elementary Matrices

Theorem

Every elementary matrices $\mathbf{E}$ are *invertible*. The *inverse* $\mathbf{E}^{-1}$ is the elementary matrix corresponding to the reverse of the row operation corresponding to $\mathbf{E}$.

(i)

$$\mathbf{I}_n \xrightarrow{R_i + cR_j} \mathbf{E} \xrightarrow{R_i - cR_j} \mathbf{I}_n \Rightarrow \mathbf{E} : R_i + cR_j, \mathbf{E}^{-1} : R_i - cR_j.$$

(ii)

$$\mathbf{I}_n \xrightarrow{R_i \leftrightarrow R_j} \mathbf{E} \xrightarrow{R_i \leftrightarrow R_j} \mathbf{I}_n \Rightarrow \mathbf{E} : R_i \leftrightarrow R_j, \mathbf{E}^{-1} : R_i \leftrightarrow R_j.$$

(iii)

$$\mathbf{I}_n \xrightarrow{cR_i} \mathbf{E} \xrightarrow{\frac{1}{c}R_i} \mathbf{I}_n \Rightarrow \mathbf{E} : cR_i, \mathbf{E}^{-1} : \frac{1}{c}R_i.$$

Example

$$1. \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_2+3R_4} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$2. \quad \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$3. \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{2R_2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Question

Find the inverse of this elementary matrix

$$\mathbf{E} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1/2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Equivalent Statements for Invertibility

Introduction

This section will be mainly proving important equivalent statements of invertibility. That is, as long as any one of these statements holds, we know that all the other statements will hold true too. The equivalent statements of invertibility is like a junction, where knowing that any of the statements opens you up to all the other statements, of which one of them might be useful in solving the problem you have at hand.

Before we get lost in a sea of theorems and proofs, we will be illustrating the statements with some examples.

Example

Consider the matrix $\mathbf{A} = \begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix}$. Reducing, we have

$$\begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix} \xrightarrow{R_1 - R_3} \xrightarrow{R_2 + R_1} \xrightarrow{R_3 + R_2} \xrightarrow{\frac{1}{2}R_3} \xrightarrow{R_2 - R_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Here we can conclude that the reduced row-echelon form of $\mathbf{A}$ is the identity matrix, and from the previous section,

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

By taking the inverse of the elementary matrices, we get

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Observe that $\mathbf{A}$ is a product of elementary matrices!

Example

Next, let us evaluate the product of the elementary matrices in the reduction of $\mathbf{A}$,

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mathbf{A} = \begin{pmatrix} 1 & 0 & -1 \\ 1/2 & 1/2 & -1 \\ 1/2 & 1/2 & 0 \end{pmatrix} \mathbf{A}.$$

Observe that if we multiply the matrix to the right of $\mathbf{A}$, we do get the identity matrix too,

$$\begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 1/2 & 1/2 & -1 \\ 1/2 & 1/2 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

This shows that $\mathbf{A}^{-1} = \begin{pmatrix} 1 & 0 & -1 \\ 1/2 & 1/2 & -1 \\ 1/2 & 1/2 & 0 \end{pmatrix}.$

Example

Now consider the linear system $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$, for some $b_1, b_2, b_3 \in \mathbb{R}$. Let's solve the equation by row reduction

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & b_1 \\ -1 & 1 & 1 & b_2 \\ 0 & -1 & 1 & b_3 \end{array} \right) \xrightarrow{R_1 - R_3} \xrightarrow{R_2 + R_1} \xrightarrow{R_3 + R_2} \xrightarrow{\frac{1}{2}R_3} \xrightarrow{R_2 - R_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & b_1 - b_3 \\ 0 & 1 & 0 & b_1/2 + b_2/2 - b_3 \\ 0 & 0 & 1 & b_1/2 + b_2/2 \end{array} \right).$$

This shows that the system is not only consistent, but have a unique solution for every $\mathbf{b}$. In fact,

$$\mathbf{A}^{-1}\mathbf{b} = \begin{pmatrix} 1 & 0 & -1 \\ 1/2 & 1/2 & -1 \\ 1/2 & 1/2 & 0 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} b_1 - b_3 \\ b_1/2 + b_2/2 - b_3 \\ b_1/2 + b_2/2 \end{pmatrix}.$$

Example

Now consider $\mathbf{A} = \begin{pmatrix} 1 & 1 & -1 \\ 1 & 0 & 0 \\ -1 & -1 & 1 \end{pmatrix}$. Reducing, we have

$$\begin{pmatrix} 1 & 1 & -1 \\ 1 & 0 & 0 \\ -1 & -1 & 1 \end{pmatrix} \xrightarrow{R_3+R_1} \xrightarrow{R_2 \leftrightarrow R_1} \xrightarrow{R_2-R_1} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}.$$

The reduced row-echelon form of $\mathbf{A}$ has both a non-pivot column and a zero row. Now consider the linear system $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$, for some $b_1, b_2, b_3 \in \mathbb{R}$. Let's solve the equation by row reduction.

$$\left(\begin{array}{ccc|c} 1 & 1 & -1 & b_1 \\ 1 & 0 & 0 & b_2 \\ -1 & -1 & 1 & b_3 \end{array} \right) \xrightarrow{R_3+R_1} \xrightarrow{R_2 \leftrightarrow R_1} \xrightarrow{R_2-R_1} \left(\begin{array}{ccc|c} 1 & 0 & 0 & b_2 \\ 0 & 1 & -1 & b_1 - b_2 \\ 0 & 0 & 0 & b_1 + b_3 \end{array} \right).$$

If $b_1 + b_3 \neq 0$, then the system is inconsistent. If $b_1 + b_3 = 0$, then the system has infinitely many solutions. Can $\mathbf{A}$ be written as a product of elementary matrices? Can the reduced row-echelon form of $\mathbf{A}$ be the identity matrix? Is $\mathbf{A}$ invertible?

Elementary Matrices and Inverse

Theorem

If $\mathbf{A} = \mathbf{E}_k \cdots \mathbf{E}_2 \mathbf{E}_1$ is a product of elementary matrices, then $\mathbf{A}$ is invertible.

Proof.

This follows from the fact that the product of invertible matrices is invertible. Hence, if a square matrix $\mathbf{A}$ can be written as a product of elementary matrices, then since elementary matrices are invertible, $\mathbf{A}$ is invertible. □

Corollary

If the reduced row-echelon form of $\mathbf{A}$ is the identity matrix, then $\mathbf{A}$ is invertible.

Proof.

By the hypothesis, $\mathbf{A}$ is row equivalent to $\mathbf{I}$. This means that there are elementary matrices $\mathbf{E}_1, \mathbf{E}_2, \dots, \mathbf{E}_k$ such that

$$\mathbf{E}_k \cdots \mathbf{E}_2 \mathbf{E}_1 \mathbf{A} = \mathbf{I}.$$

By premultiplying the inverse of the elementary matrices, we have

$$\mathbf{A} = \mathbf{E}_1^{-1} \mathbf{E}_2^{-1} \cdots \mathbf{E}_k^{-1} \mathbf{I} = \mathbf{E}_1^{-1} \mathbf{E}_2^{-1} \cdots \mathbf{E}_k^{-1}.$$

Equivalent Statements for Invertibility

Theorem

A square matrix $\mathbf{A}$ is invertible if and only if the homogeneous system $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution.

Proof.

($\Rightarrow$) We have shown that if $\mathbf{A}$ is invertible, then the homogeneous system $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution.

($\Leftarrow$) Now suppose the homogeneous system $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution. This necessitates the reduced row-echelon form of $\mathbf{A}$ to be the identity matrix. For otherwise, the reduced row-echelon form must have a non-pivot column, which then implies that the homogeneous system $\mathbf{Ax} = \mathbf{0}$ has infinitely many solutions, a contradiction. Hence, by the previous corollary, $\mathbf{A}$ is invertible. $\square$

The proof also shows that if $\mathbf{A}$ is invertible, then its reduced row-echelon form is the identity matrix.

Theorem

A square matrix $\mathbf{A}$ is invertible if and only if its reduced row-echelon form is the identity matrix.

Theorem

A square matrix $\mathbf{A}$ is invertible if and only if it is a product of elementary matrices.

The proofs of theorems are left as exercises.

Left and Right Inverses

Definition

Let $\mathbf{A}$ be a $n \times m$ matrix.

- (i) A $m \times n$ matrix $\mathbf{B}$ is said to be a left inverse of $\mathbf{A}$ if $\mathbf{BA} = \mathbf{I}_m$, where $\mathbf{I}_m$ is the $m \times m$ identity matrix.
- (ii) A $m \times n$ matrix $\mathbf{B}$ is said to be a right inverse of $\mathbf{A}$ if $\mathbf{AB} = \mathbf{I}_n$, where $\mathbf{I}_n$ is the $n \times n$ identity matrix.

Observe that $\mathbf{B}$ is a **left inverse** of $\mathbf{A}$ if and only if $\mathbf{A}$ is a **right inverse** of $\mathbf{B}$.

Example

1. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ is a left inverse of $\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$; $\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$ is a right inverse of $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$.

2. The inverse of a square matrix $\mathbf{A}$ is both a (the) left and right inverse, $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I} = \mathbf{AA}^{-1}$.

Equivalent Statements for Invertibility

Theorem

A square matrix $\mathbf{A}$ is invertible if and only if it has a left inverse.

Proof.

($\Rightarrow$) An inverse is a left inverse.

($\Leftarrow$) Suppose now $\mathbf{B}$ is a left inverse of $\mathbf{A}$, $\mathbf{BA} = \mathbf{I}$. Consider now the homogeneous system $\mathbf{Ax} = \mathbf{0}$. Let $\mathbf{u}$ be a solution, $\mathbf{Au} = \mathbf{0}$. Then premultiplying by the left inverse $\mathbf{B}$,

$$\mathbf{0} = \mathbf{B}(\mathbf{0}) = \mathbf{B}(\mathbf{Au}) = (\mathbf{BA})\mathbf{u} = \mathbf{I}\mathbf{u} = \mathbf{u}$$

tells us that $\mathbf{u} = \mathbf{0}$, that is, the homogeneous system has only the trivial solution. Hence, $\mathbf{A}$ is invertible. □

Equivalent Statements for Invertibility

Theorem

A square matrix $\mathbf{A}$ is invertible if and only if it has a right inverse.

Proof.

Recall that $\mathbf{A}$ is invertible if and only if $\mathbf{A}^T$ is. Let $\mathbf{B}$ be a right inverse of $\mathbf{A}$, $\mathbf{AB} = \mathbf{I}$. Then by taking the transpose, and observing that the identity matrix is symmetric, we have $\mathbf{B}^T\mathbf{A}^T = \mathbf{I}$. This shows that $\mathbf{B}^T$ is a left inverse of $\mathbf{A}^T$. Therefore, by the previous theorem, $\mathbf{A}^T$ is invertible, which thus proves that $\mathbf{A}$ is invertible. □

Equivalent Statement for Invertibility

Theorem

A square matrix $\mathbf{A}$ is invertible if and only if $\mathbf{Ax} = \mathbf{b}$ has a unique solution for all $\mathbf{b}$.

Proof.

($\Rightarrow$) We have shown that if $\mathbf{A}$ is invertible, then $\mathbf{Ax} = \mathbf{b}$ has a unique solution for all $\mathbf{b}$.

($\Leftarrow$) Now suppose $\mathbf{Ax} = \mathbf{b}$ is consistent for all $\mathbf{b}$. In particular, $\mathbf{Ax} = \mathbf{e}_i$ is consistent, where $\mathbf{e}_i$ is the i -th column of the identity matrix, for $i = 1, \dots, n$, where n is the order of $\mathbf{A}$. Let $\mathbf{b}_i$ be a solution to $\mathbf{Ax} = \mathbf{e}_i$. Let $\mathbf{B} = (\mathbf{b}_1 \ \mathbf{b}_2 \ \cdots \ \mathbf{b}_n)$. Then, by block multiplication

$$\mathbf{AB} = \mathbf{A}(\mathbf{b}_1 \ \mathbf{b}_2 \ \cdots \ \mathbf{b}_n) = (\mathbf{Ab}_1 \ \mathbf{Ab}_2 \ \cdots \ \mathbf{Ab}_n) = (\mathbf{e}_1 \ \mathbf{e}_2 \ \cdots \ \mathbf{e}_n) = \mathbf{I},$$

which shows that $\mathbf{B}$ is a right inverse of $\mathbf{A}$. Thus, by the previous theorem, $\mathbf{A}$ is invertible. □

Algorithm for Finding Inverse

Thus, we will now formally introduce the algorithm to testing if a matrix is invertible, and finding its inverse if it is invertible.

Let $\mathbf{A}$ be a $n \times n$ matrix.

Step 1: Form the $n \times 2n$ (augmented) matrix $(\mathbf{A} \mid \mathbf{I}_n)$.

Step 2: Reduce the matrix $(\mathbf{A} \mid \mathbf{I}) \longrightarrow (\mathbf{R} \mid \mathbf{B})$ to its REF or RREF.

Step 3: If RREF $\mathbf{R} \neq \mathbf{I}$ or REF has a zero row, then $\mathbf{A}$ is not invertible. If RREF $\mathbf{R} = \mathbf{I}$ or REF has no zero row, $\mathbf{A}$ is invertible with inverse $\mathbf{A}^{-1} = \mathbf{B}$.

Question

Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times m$ matrices. Show that $\mathbf{A}$ and $\mathbf{B}$ are row equivalent if and only if $\mathbf{B} = \mathbf{PA}$ for some invertible $n \times n$ matrix $\mathbf{P}$.

Equivalent Statements of Invertibility

Theorem (Equivalent statements of invertibility)

Let $\mathbf{A}$ be a *square* matrix of order n . The following statements are equivalent.

- (i) $\mathbf{A}$ is *invertible*.
- (ii) $\mathbf{A}^T$ is *invertible*.
- (iii) (*left inverse*) There is a matrix $\mathbf{B}$ such that $\mathbf{BA} = \mathbf{I}$.
- (iv) (*right inverse*) There is a matrix $\mathbf{B}$ such that $\mathbf{AB} = \mathbf{I}$.
- (v) The *reduced row-echelon form* of $\mathbf{A}$ is the *identity matrix*.
- (vi) $\mathbf{A}$ can be expressed as a *product* of *elementary matrices*.
- (vii) The *homogeneous system* $\mathbf{Ax} = \mathbf{0}$ has *only the trivial solution*.
- (viii) For *any* $\mathbf{b}$, the system $\mathbf{Ax} = \mathbf{b}$ has a *unique solution*.

Determinant by Cofactor Expansion

Order 1 and 2 Square Matrices

We will define the determinant of $\mathbf{A}$ of order n , denoted as $\det(\mathbf{A})$, or $|\mathbf{A}|$, by induction.

1. For $n = 1$, $\mathbf{A} = (a)$, $\det(\mathbf{A}) = a$.
2. For $n = 2$, $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\det(\mathbf{A}) = ad - bc$.

Example

(i) $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}$, $\det(\mathbf{A}) = (1)(1) - (2)(3) = -5$.

(ii) $\mathbf{A} = \begin{pmatrix} 3 & -1 \\ 5 & 1 \end{pmatrix}$, $\det(\mathbf{A}) = (3)(1) - (-1)(5) = 8$.

Inductive Step: Matrix Minor

Suppose we have defined the determinant of all square matrices of order $\leq n - 1$. Let $\mathbf{A}$ be a square matrix of order n .

Define $\mathbf{M}_{ij}$, called the (i, j) matrix minor of $\mathbf{A}$, to be the matrix obtained from $\mathbf{A}$ by deleting the i -th row and j -th column.

Example

$$\mathbf{A} = \begin{pmatrix} 5 & 1 & 2 & -1 \\ -1 & -3 & 1 & 3 \\ 3 & 8 & 2 & 1 \\ 2 & 0 & 1 & 11 \end{pmatrix}$$

$$\mathbf{M}_{11} = \begin{pmatrix} -3 & 1 & 3 \\ 8 & 2 & 1 \\ 0 & 1 & 11 \end{pmatrix}, \quad \mathbf{M}_{12} = \begin{pmatrix} -1 & 1 & 3 \\ 3 & 2 & 1 \\ 2 & 1 & 11 \end{pmatrix}, \quad \mathbf{M}_{23} = \begin{pmatrix} 5 & 1 & -1 \\ 3 & 8 & 1 \\ 2 & 0 & 11 \end{pmatrix}, \quad \mathbf{M}_{43} = \begin{pmatrix} 5 & 1 & -1 \\ -1 & -3 & 3 \\ 3 & 8 & 1 \end{pmatrix}$$

Inductive Step: Cofactor

The (i,j) -cofactor of $\mathbf{A}$, denoted as A_{ij} , is the (real) number given by

$$A_{ij} = (-1)^{i+j} \det(\mathbf{M}_{ij}).$$

Take note of the sign of the (i,j) -entry, $(-1)^{i+j}$. Here's a visualization of the sign of the entries of the matrix

$$\begin{pmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \ddots & \end{pmatrix}.$$

Example

$$\begin{pmatrix} 1 & 2 & -1 \\ -1 & 1 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

A_{11}

A_{12}

A_{13}

A_{21}

A_{22}

A_{23}

A_{31}

A_{32}

A_{33}

Example

$$\begin{pmatrix} 1 & 2 & -1 \\ -1 & 1 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$\begin{aligned} A_{11} &= \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} \\ &= (1)(1) - (2)(3) \\ &= -5 \end{aligned}$$

$$\begin{aligned} A_{12} &= -\begin{vmatrix} -1 & 3 \\ 3 & 1 \end{vmatrix} \\ &= -(-1)(1) + (3)(3) \\ &= 10 \end{aligned}$$

$$\begin{aligned} A_{13} &= \begin{vmatrix} -1 & 1 \\ 3 & 2 \end{vmatrix} \\ &= (-1)(2) - (1)(3) \\ &= -5 \end{aligned}$$

$$\begin{aligned} A_{21} &= -\begin{vmatrix} 2 & -1 \\ 2 & 1 \end{vmatrix} \\ &= -(2)(1) + (-1)(2) \\ &= -4 \end{aligned}$$

$$\begin{aligned} A_{22} &= \begin{vmatrix} 1 & -1 \\ 3 & 1 \end{vmatrix} \\ &= (1)(1) - (-1)(3) \\ &= 4 \end{aligned}$$

$$\begin{aligned} A_{23} &= -\begin{vmatrix} 1 & 2 \\ 3 & 2 \end{vmatrix} \\ &= -(1)(2) + (2)(3) \\ &= 4 \end{aligned}$$

$$\begin{aligned} A_{31} &= \begin{vmatrix} 2 & -1 \\ 1 & 3 \end{vmatrix} \\ &= (2)(3) - (-1)(1) \\ &= 7 \end{aligned}$$

$$\begin{aligned} A_{32} &= -\begin{vmatrix} 1 & -1 \\ -1 & 3 \end{vmatrix} \\ &= -(1)(3) + (-1)(-1) \\ &= -2 \end{aligned}$$

$$\begin{aligned} A_{33} &= \begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix} \\ &= (1)(1) - (2)(-1) \\ &= 3 \end{aligned}$$

Determinant by Cofactor Expansion

The determinant of $\mathbf{A}$ is defined to be

$$\det(\mathbf{A}) = a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in} = \sum_{k=1}^n a_{ik}A_{ik} \quad (1)$$

$$= a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj} = \sum_{k=1}^n a_{kj}A_{kj} \quad (2)$$

This is called the cofactor expansion along $\begin{cases} \text{row } i & (1) \\ \text{column } j & (2) \end{cases}$.

Remark

The above is both a theorem and a definition. The theorem states that evaluating the cofactor expansion along any row or column produces the same result. Hence, we may define the determinant to be the cofactor expansion along any rows or columns.

Property of Determinant

Theorem (Determinant is invariant under transpose)

Let $\mathbf{A}$ be a *square* matrix. Then the *determinant* of $\mathbf{A}$ is *equal* to the *determinant* of $\mathbf{A}^T$,

$$\det(\mathbf{A}) = \det(\mathbf{A}^T).$$

The idea of the proof is that taking the cofactor expansion of $\mathbf{A}$ along column 1 is equal to the cofactor expansion of $\mathbf{A}^T$ along row 1. But the first produces $\det(\mathbf{A})$, while the latter $\det(\mathbf{A}^T)$.

Order 3 Matrices

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix} = aei - afh - bdi + bfg + cdh - ceg.$$

$$\begin{vmatrix} a & b & c & a & b \\ d & e & f & d & e \\ g & h & i & g & h \end{vmatrix}$$

Example

$$\begin{vmatrix} 0 & 1 & 1 \\ 2 & 1 & 0 \\ 1 & 2 & 3 \end{vmatrix} = (0)(1)(3) + (1)(0)(1) + (1)(2)(2) - (1)(1)(1) - (1)(2)(3) - (0)(0)(2) = -3$$

Example

$$1. \begin{vmatrix} 1 & 3 & 1 \\ 2 & 1 & 0 \\ 1 & 2 & 3 \end{vmatrix} = (1)(1)(3) + (3)(0)(1) + (1)(2)(2) - (1)(1)(1) - (3)(2)(3) - (1)(0)(2) = -12$$

$$2. \begin{vmatrix} 2 & 1 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \end{vmatrix} = 2 \begin{vmatrix} 0 & 1 & 1 \\ 2 & 1 & 0 \\ 1 & 2 & 3 \end{vmatrix} - \begin{vmatrix} 1 & 3 & 1 \\ 2 & 1 & 0 \\ 1 & 2 & 3 \end{vmatrix} = 2(-3) - (-12) = 6 \text{ (cofactor expansion along column 1)}$$

$$3. \begin{vmatrix} 1 & 5 & 1 & 2 \\ 0 & 2 & 6 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{vmatrix} = (1) \begin{vmatrix} 2 & 6 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{vmatrix} = (1)(2) \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = (1)(2)(1)(1) = 2. \text{ Observe that the determinant of a triangular matrix is the product of the diagonal entries.}$$

$$4. \begin{vmatrix} 1 & 1 & 0 & 1 \\ 0 & 2 & 0 & 1 \\ 1 & 1 & 2 & 3 \\ 0 & 2 & 0 & 2 \end{vmatrix} = 2 \begin{vmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 2 & 2 \end{vmatrix} = (2)(1) \begin{vmatrix} 2 & 1 \\ 2 & 2 \end{vmatrix} = 2(4 - 2) = 4 \text{ (cofactor expansion along column 3, then along column 1)}$$

Question

What is the determinant of the following matrix?

$$\begin{pmatrix} 1 & 1 & 3 & 0 & 5 & -2 \\ 2 & 2 & 0 & 0 & 1 & 0 \\ -1 & 7 & 2 & 0 & 0 & 9 \\ 0 & 1 & 0 & 0 & 1 & 2 \\ 1 & 1 & 0 & 0 & 2 & 1 \\ 1 & -5 & 7 & 0 & 8 & 0 \end{pmatrix}$$

Determinant of Triangular Matrices

Theorem (Determinant of a triangular matrix is the product of diagonal entries)

If $\mathbf{A} = (a_{ij})_n$ is a triangular matrix, then

$$\det(\mathbf{A}) = a_{11}a_{22} \cdots a_{nn} = \prod_{k=1}^n a_{kk}.$$

Sketch of proof.

Upper triangular matrix, continuously cofactor expand along first column,

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} & a_{24} \\ 0 & a_{33} & a_{34} \\ 0 & 0 & a_{44} \end{vmatrix} = a_{11}a_{22} \begin{vmatrix} a_{33} & a_{34} \\ 0 & a_{44} \end{vmatrix} = a_{11}a_{22}a_{33}a_{44}.$$

Lower triangular matrix, continuously cofactor expand along the first row,

$$\begin{vmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & 0 & 0 \\ a_{32} & a_{33} & 0 \\ a_{42} & a_{43} & a_{44} \end{vmatrix} = a_{11}a_{22} \begin{vmatrix} a_{33} & 0 \\ a_{43} & a_{44} \end{vmatrix} = a_{11}a_{22}a_{33}a_{44}.$$



Example

$$\begin{vmatrix} 1 & 5 & 1 & 2 \\ 0 & 2 & 6 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{vmatrix} = (1)(2)(1)(1) = 2.$$

Upshot: Computation of determinant of triangular matrices is easy!

But is it useful? Majority of the square matrices are not triangular. But, every matrix is related to some triangular matrix!

Question

1. Fill in the blanks. Every square matrix is _____ to a triangular matrix.
2. Suppose **A** and **B** are row equivalent. How are the determinants of **A** and **B** related?

Determinant by Reduction

Determinant and Elementary Row Operations

$R_i + aR_j$

$$\det(\mathbf{B}) = \begin{vmatrix} 1 & 5 & 1 & 2 \\ 0 & 2 & 6 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 2$$

$$\mathbf{A} = \begin{pmatrix} 1 & 5 & 1 & 2 \\ 0 & 2 & 6 & 3 \\ 3 & 15 & 4 & 8 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_3 - 3R_1} \begin{pmatrix} 1 & 5 & 1 & 2 \\ 0 & 2 & 6 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \mathbf{B}$$

$$\det(\mathbf{A}) = \begin{vmatrix} 1 & 5 & 1 & 2 \\ 0 & 2 & 6 & 3 \\ 3 & 15 & 4 & 8 \\ 0 & 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 5 & 1 \\ 0 & 2 & 6 \\ 3 & 15 & 4 \end{vmatrix} = \begin{vmatrix} 2 & 6 \\ 15 & 4 \end{vmatrix} + 3 \begin{vmatrix} 5 & 1 \\ 2 & 6 \end{vmatrix} = 2.$$

$$\det(\mathbf{B}) = \det(\mathbf{A}).$$

Determinant and Elementary Row Operations

cR_i

$$\det(\mathbf{B}) = \begin{vmatrix} 1 & 5 & 1 & 2 \\ 0 & 2 & 6 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 2$$

$$\mathbf{A} = \begin{pmatrix} 1 & 5 & 1 & 2 \\ 0 & 4 & 12 & 6 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{\frac{1}{2}R_2} \begin{pmatrix} 1 & 5 & 1 & 2 \\ 0 & 2 & 6 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \mathbf{B}$$

$$\det(\mathbf{A}) = \begin{vmatrix} 1 & 5 & 1 & 2 \\ 0 & 4 & 12 & 6 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 4.$$

$$\det(\mathbf{B}) = \frac{1}{2} \det(\mathbf{A}).$$

Determinant and Elementary Row Operations

$$R_i \leftrightarrow R_j$$

$$\det(\mathbf{B}) = \begin{vmatrix} 1 & 5 & 1 & 2 \\ 0 & 2 & 6 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 2$$

$$\mathbf{A} = \begin{pmatrix} 1 & 5 & 1 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 2 & 6 & 3 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_4} \begin{pmatrix} 1 & 5 & 1 & 2 \\ 0 & 2 & 6 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \mathbf{B}$$

$$\det(\mathbf{A}) = \begin{vmatrix} 1 & 5 & 1 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 2 & 6 & 3 \end{vmatrix} = \begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 2 & 6 & 3 \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 2 & 6 \end{vmatrix} = -2.$$

$$\det(\mathbf{B}) = -\det(\mathbf{A}).$$

Determinant and Elementary Row Operations

Theorem

Let $\mathbf{A}$ be a $n \times n$ square matrix. Suppose $\mathbf{B}$ is obtained from $\mathbf{A}$ via a single elementary row operation. Then the determinant of $\mathbf{B}$ is obtained as such.

$\mathbf{A} \xrightarrow{R_i + aR_j} \mathbf{B}$	$\det(\mathbf{B}) = \det(\mathbf{A})$
$\mathbf{A} \xrightarrow{cR_i} \mathbf{B}$	$\det(\mathbf{B}) = c \det(\mathbf{A})$
$\mathbf{A} \xrightarrow{R_i \leftrightarrow R_j} \mathbf{B}$	$\det(\mathbf{B}) = -\det(\mathbf{A})$

Corollary

The determinant of an elementary matrix $\mathbf{E}$ is given as such.

- (i) If $\mathbf{E}$ corresponds to $R_i + aR_j$, then $\det(\mathbf{E}) = 1$.
- (ii) If $\mathbf{E}$ corresponds to cR_j , then $\det(\mathbf{E}) = c$.
- (iii) If $\mathbf{E}$ corresponds to $R_i \leftrightarrow R_j$, then $\det(\mathbf{E}) = -1$.

Determinant of Row Equivalent Matrices

Theorem

Let $\mathbf{A}$ and $\mathbf{R}$ be square matrices such that

$$\mathbf{R} = \mathbf{E}_k \cdots \mathbf{E}_2 \mathbf{E}_1 \mathbf{A}$$

for some elementary matrices $\mathbf{E}_1, \mathbf{E}_2, \dots, \mathbf{E}_k$. Then

$$\det(\mathbf{R}) = \det(\mathbf{E}_k) \cdots \det(\mathbf{E}_2) \det(\mathbf{E}_1) \det(\mathbf{A}).$$

Proof.

This follows immediately from the previous theorem and corollary.



Remark

Note that the determinant of an elementary matrix is nonzero. This means that the determinant of $\mathbf{R}$ could be computed from the determinant of $\mathbf{A}$ and vice versa.

Determinant of Row Equivalent Matrices

Corollary

Let $\mathbf{A}$ be a $n \times n$ square matrix. Suppose $\mathbf{A} \xrightarrow{r_1} \xrightarrow{r_2} \cdots \xrightarrow{r_k} \mathbf{R} = \begin{pmatrix} d_1 & * & \cdots & * \\ 0 & d_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}$, where $\mathbf{R}$ is a

row-echelon form of $\mathbf{A}$. Let $\mathbf{E}_i$ be the elementary matrix corresponding to the elementary row operation r_i , for $i = 1, \dots, k$. Then

$$\det(\mathbf{A}) = \frac{d_1 d_2 \cdots d_n}{\det(\mathbf{E}_k) \cdots \det(\mathbf{E}_2) \det(\mathbf{E}_1)}.$$

Proof.

Using the previous theorem and the fact that the determinant of a triangular matrix is the product its diagonal entries,

$$d_1 d_2 \cdots d_n = \det(\mathbf{R}) = \det(\mathbf{E}_k) \cdots \det(\mathbf{E}_2) \det(\mathbf{E}_1) \det(\mathbf{A}).$$

Hence,

$$\det(\mathbf{A}) = \frac{d_1 d_2 \cdots d_n}{\det(\mathbf{E}_k) \cdots \det(\mathbf{E}_2) \det(\mathbf{E}_1)}.$$

Example

$$\begin{aligned} \mathbf{A} = \begin{pmatrix} 2 & 1 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \end{pmatrix} &\xrightarrow[\mathbf{E}_2: R_3 - 2R_4]{\mathbf{E}_1: R_1 - 2R_2} \begin{pmatrix} 0 & 1 & 1 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & -3 & -6 \\ 0 & 1 & 2 & 3 \end{pmatrix} \xrightarrow[\mathbf{E}_4: -\frac{1}{3}R_3]{\mathbf{E}_3: R_4 - R_1} \\ \begin{pmatrix} 0 & 1 & 1 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 4 \end{pmatrix} &\xrightarrow{\mathbf{E}_5: R_4 - R_3} \begin{pmatrix} 0 & 1 & 1 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 2 \end{pmatrix} \xrightarrow{\mathbf{E}_6: R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 2 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \Rightarrow 2 &= \det(\mathbf{E}_6) \det(\mathbf{E}_5) \det(\mathbf{E}_4) \det(\mathbf{E}_3) \det(\mathbf{E}_2) \det(\mathbf{E}_1) \det(\mathbf{A}) \\ &= (-1)(1) \left(-\frac{1}{3}\right) (1)(1)(1) \det(\mathbf{A}) \end{aligned}$$

$$\Rightarrow \det(\mathbf{A}) = 6$$

Properties of Determinant

Determinant of Product of Matrices

Theorem

Let $\mathbf{A}$ and $\mathbf{B}$ be square matrices of the same size. Then

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B}).$$

By induction, we get

$$\det(\mathbf{A}_1 \mathbf{A}_2 \cdots \mathbf{A}_k) = \det(\mathbf{A}_1) \det(\mathbf{A}_2) \cdots \det(\mathbf{A}_k).$$

Determinant and Invertibility

Theorem

A *square matrix* $\mathbf{A}$ is *invertible* if and only if $\det(\mathbf{A}) \neq 0$. In this case, the determinant of the inverse is

$$\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}.$$

Sketch of Proof.

Assuming $\mathbf{A}$ is invertible. Since the identity matrix is a triangular matrix, the determinant is just the product of the diagonal entries, which is equal to 1. Hence,

$$1 = \det(\mathbf{I}) = \det(\mathbf{A}^{-1}\mathbf{A}) = \det(\mathbf{A}^{-1})\det(\mathbf{A})$$

$$\Rightarrow \det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}.$$



Equivalent Statements for Invertibility

Let $\mathbf{A}$ be a square matrix of order n . The following statements are equivalent.

- (i) $\mathbf{A}$ is invertible.
- (ii) $\mathbf{A}^T$ is invertible.
- (iii) (left inverse) There is a matrix $\mathbf{B}$ such that $\mathbf{BA} = \mathbf{I}$.
- (iv) (right inverse) There is a matrix $\mathbf{B}$ such that $\mathbf{AB} = \mathbf{I}$.
- (v) The reduced row-echelon form of $\mathbf{A}$ is the identity matrix.
- (vi) $\mathbf{A}$ can be expressed as a product of elementary matrices.
- (vii) The homogeneous system $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution.
- (viii) For any $\mathbf{b}$, the system $\mathbf{Ax} = \mathbf{b}$ has a unique solution.
- (ix) The determinant of $\mathbf{A}$ is nonzero, $\det(\mathbf{A}) \neq 0$.

Determinant of Scalar Multiplication

Theorem

For any square matrix $\mathbf{A}$ of order n and scalar $c \in \mathbb{R}$,

$$\det(c\mathbf{A}) = c^n \det(\mathbf{A}).$$

Proof.

Observe that scalar multiplication is equivalent to matrix multiplication by scalar matrix,

$$c\mathbf{A} = (c\mathbf{I})\mathbf{A} = \begin{pmatrix} c & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & c \end{pmatrix} \mathbf{A}. \text{ Hence,}$$

$$\det(c\mathbf{A}) = \det((c\mathbf{I})\mathbf{A}) = \begin{vmatrix} c & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & c \end{vmatrix} \det(\mathbf{A}) = c^n \det(\mathbf{A}).$$



Properties of Determinant

In summary.

Theorem

1. $\det(\mathbf{A}) = \det(\mathbf{A}^T)$

2. $\det(\mathbf{A}_1 \mathbf{A}_2 \cdots \mathbf{A}_k) = \det(\mathbf{A}_1) \det(\mathbf{A}_2) \cdots \det(\mathbf{A}_k)$

3. If $\mathbf{A}$ is *invertible*, then

$$\det(\mathbf{A}^{-1}) = \det(\mathbf{A})^{-1}.$$

4. If $\mathbf{A}$ is an order n square matrix and c is a constant, then

$$\det(c\mathbf{A}) = c^n \det(\mathbf{A}).$$

Example

$$\text{Let } \mathbf{A} = \begin{pmatrix} 1 & 5 & 1 & 2 \\ 0 & 2 & 6 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

$$\mathbf{A} \xrightarrow{R_4 - R_3} \begin{pmatrix} 1 & 5 & 1 & 2 \\ 0 & 2 & 6 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

So $\det(\mathbf{A}) = -2$ and $\det(\mathbf{B}) = 3$.

1. $\det(3\mathbf{A}^T) =$

2. $\det(3\mathbf{A}\mathbf{B}^{-1}) =$

3. $\det((3\mathbf{B})^{-1}) =$

Example

$$\text{Let } \mathbf{A} = \begin{pmatrix} 1 & 5 & 1 & 2 \\ 0 & 2 & 6 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

$$\mathbf{A} \xrightarrow{R_4 - R_3} \begin{pmatrix} 1 & 5 & 1 & 2 \\ 0 & 2 & 6 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

So $\det(\mathbf{A}) = -2$ and $\det(\mathbf{B}) = 3$.

1. $\det(3\mathbf{A}^T) = 3^4 \det(\mathbf{A}^T) = 3^4 \det(\mathbf{A}) = -162$

2. $\det(3\mathbf{AB}^{-1}) = 3^4 \det(\mathbf{AB}^{-1}) = 3^4 \det(\mathbf{A}) \det(\mathbf{B}^{-1}) = 3^4 \det(\mathbf{A}) \frac{1}{\det(\mathbf{B})} = -54$

3. $\det((3\mathbf{B})^{-1}) = \frac{1}{\det(3\mathbf{B})} = \frac{1}{3^4 \det(\mathbf{B})} = \frac{1}{3^5} = \frac{1}{243}.$